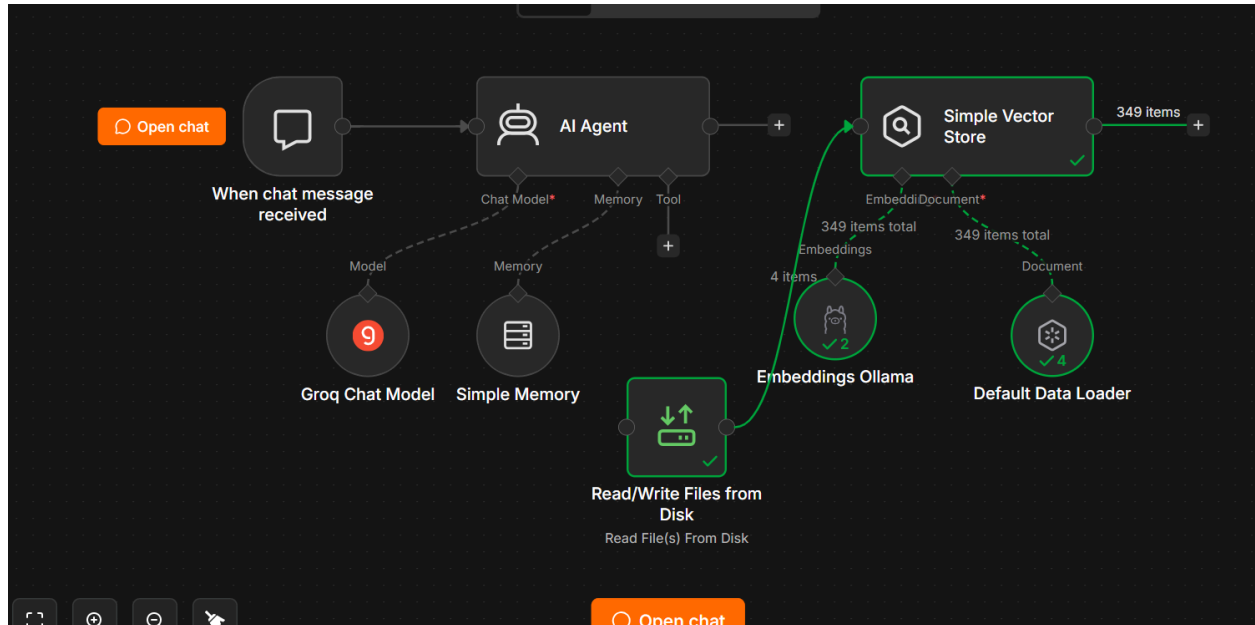


CyberON: Cybersecurity assistant chatbot

Inserting documents into the Simple vector store database



Simple Vector Store

INPUT | Schema | Table | JSON | Binary | Parameters | Settings | Execute step

Read/Write Files from Disk | 4 items

1

data

File Name: CYBER SECURITY (R18A0521).pdf
Directory: /files
File Extension: pdf
Mime Type: application/pdf
File Size: 1.72 MB

Download

2

Tip: Get a feel for vector stores in n8n with our [RAG starter template](#)

Operation Mode
Insert Documents

Memory Key
From list | cybersecurity_knowledge

Embedding Batch Size
200

☐ Clear Store

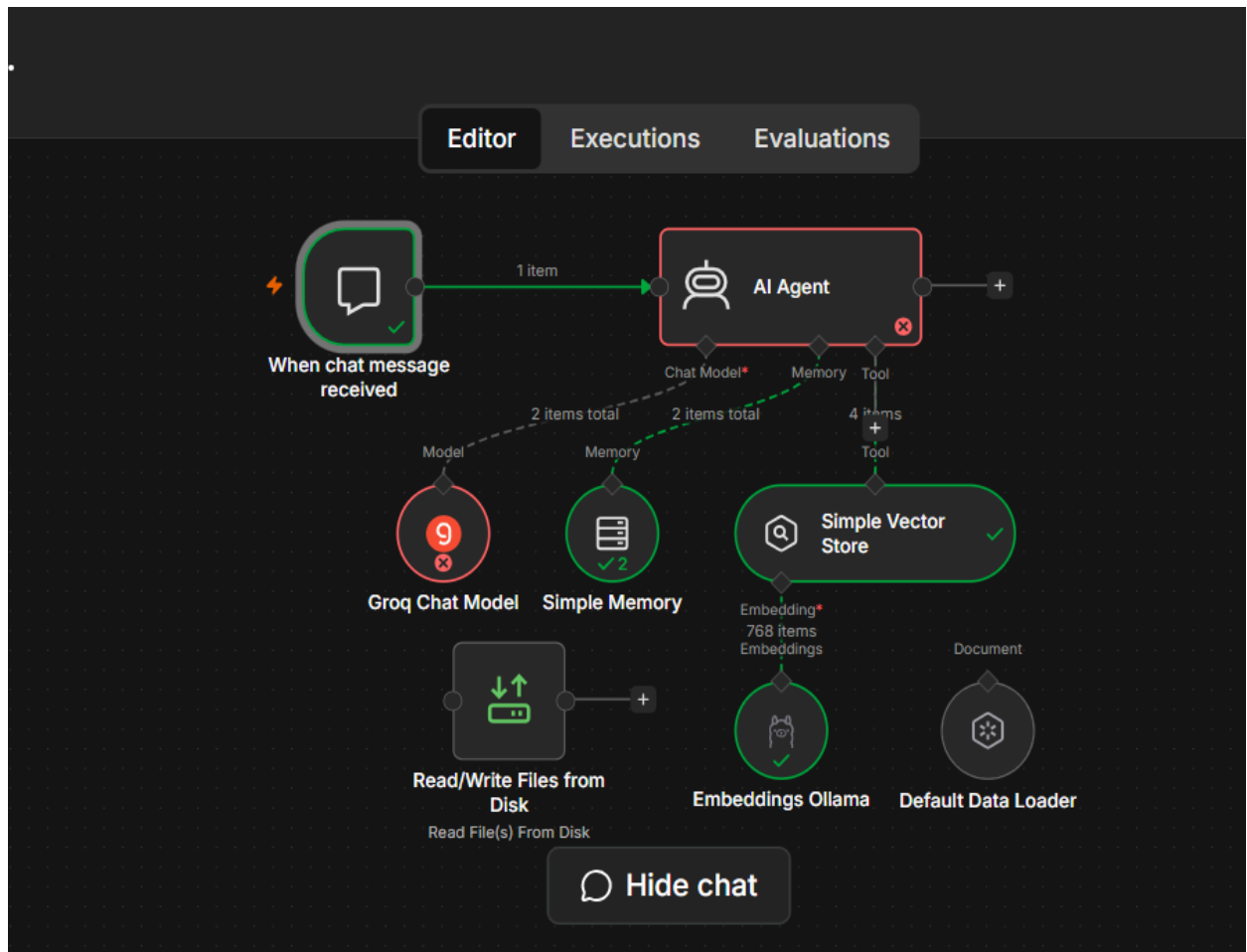
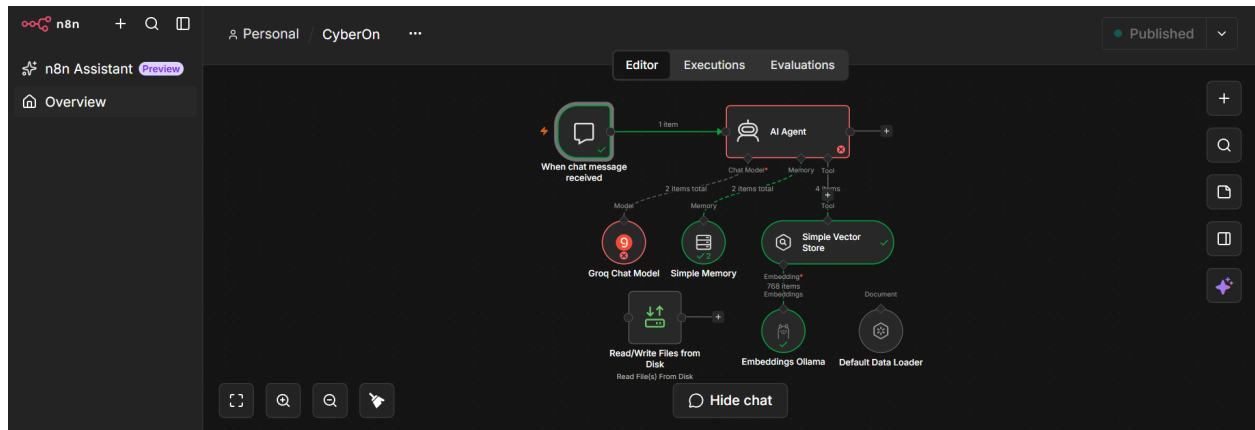
For experimental use only: Data is stored in memory and will be lost if n8n restarts. Data may also be cleared if available memory gets low, and is accessible to all users of this instance. [More info](#)

349 items

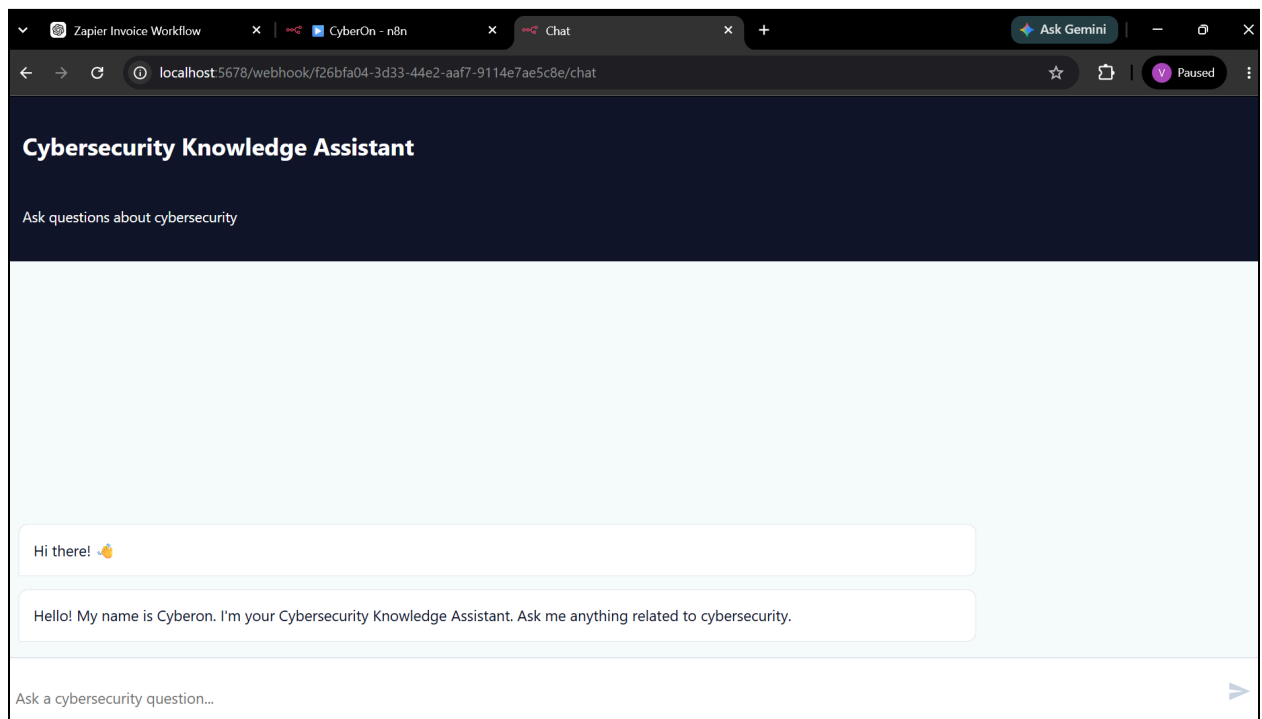
metadata	pageContent
source : blob blobType : application/pdf pdf info PDFFormatVersion : 1.5 Language : en-US EncryptFilterName : null IsLinearized : false IsAcroFormPresent : false IsXFAPresent : false IsCollectionPresent : false IsSignaturesPresent : false Author : GOJE ROOPA Creator : Microsoft® Word 2016 CreationDate : D:20210215094257+00'00' Producer : www.ilovepdf.com	CYBER SECUR NOTES\ON\ SECURITY\ SEM (R18)\n INFORMATION COLLEGE OF I TECHNOLOGY Govt. of India) of UGC ACT 1 Hyderabad, A NBA & NAAC - Certified)\n Hakimpet), Se State, India

Workflow Architecture

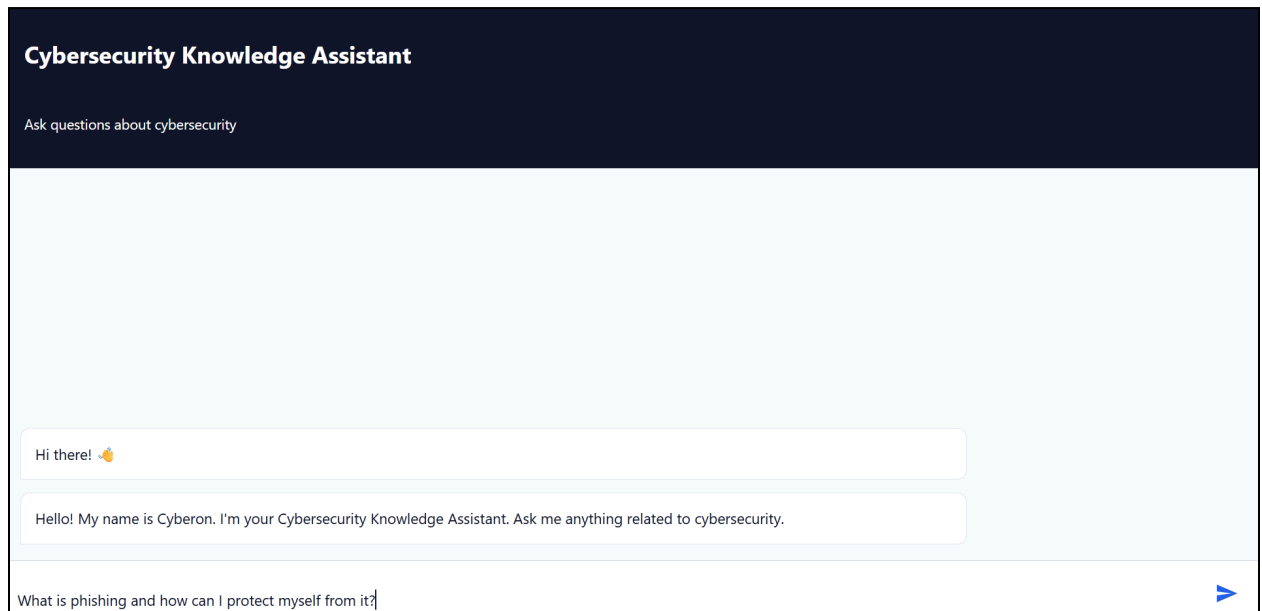
(the nodes are red as, for a brief moment of time the token limit of the Groq model had reached)
Otherwise the chatbot is fully functionally working.



Chatbot interface



First query as a user



Response

Cybersecurity Knowledge Assistant

Ask questions about cybersecurity

What is phishing and how can I protect myself from it?

What is phishing?

Phishing is a **social-engineering attack** in which an adversary pretends to be a trusted person, organization, or service and tries to convince the victim to:

- **Reveal sensitive information** (usernames, passwords, credit-card numbers, personal data).
- **Perform an unsafe action** (click a malicious link, open an infected attachment, approve a money transfer, install software, etc.).

The attack is typically delivered via **email**, but it can also appear in instant-messaging, SMS (*smishing*), voice calls (*vishing*), or social-media messages. The goal is to **steal credentials, install malware, or obtain money**.

Typical phishing workflow

1. **Preparation** – The attacker registers a look-alike domain, spoofs a “From” address, and crafts a convincing message that mimics a

Ask a cybersecurity question...

More technical query

Cybersecurity Knowledge Assistant

Ask questions about cybersecurity

What is the principle of least privilege, and where is it commonly used?

The Principle of Least Privilege (PoLP)

Definition

The Principle of Least Privilege (PoLP) states that any user, process, service, or system component should be granted only the minimum rights and permissions necessary to perform its required tasks, and nothing more. When those rights are no longer needed, they should be revoked or reduced.

Aspect	What it means
Scope	Applies to people (users, administrators), software (applications, services, scripts), and hardware (devices).
Granularity	Permissions are granted at the most specific level possible (e.g., a single file, a single API call, a single database row).
Temporal dimension	Rights can be time-bound (e.g., “only for the duration of a project”) and must be reviewed regularly .

Ask a cybersecurity question...

Bottom line

The **Principle of Least Privilege** is a fundamental security control that limits the damage an attacker—or an accidental mistake—can cause. It is applied across **operating systems, cloud IAM, databases, applications, networks, privileged-access tools, and even physical security**. By systematically identifying the exact permissions needed, assigning them narrowly, and continuously reviewing them, organizations dramatically reduce risk and meet many regulatory requirements.

Ask a cybersecurity question...

Follow up query to check memory

Cybersecurity Knowledge Assistant

Ask questions about cybersecurity

Can you give me a practical example of how this would work in an organisation?

Below is a **step-by-step, end-to-end** illustration of how the **Principle of Least Privilege (PoLP)** can be put into practice in a mid-size enterprise that runs a hybrid environment (on-prem AD + Azure AD + AWS).

The scenario follows a **new finance analyst** who needs to prepare monthly reports that draw data from an on-prem SQL Server, a cloud-based data-lake (AWS S3), and a Power-BI dashboard hosted in Azure.

1 Business Requirement (What the user must do)

Task	Systems involved	Data needed
View financial transactions stored in the on-prem SQL Server FinanceDB.Transactions table	On-prem Windows server running SQL Server 2019	Read-only
Pull raw sales CSV files from an S3 bucket	AWS S3	Read-only

Ask a cybersecurity question...

6 Summary Diagram (textual)

```
[Finance Analyst (AD user)]
|
|--- AD Group: Finance-Analyst (standard user)
|
|--- On-Prem Resources
|   |--- SQL Server: SELECT only on FinanceDB.Transactions
|   |--- File Server: Write only to \\fileserver\Finance\Reports
|
|--- Cloud Resources
|   |--- AWS: Assume Role FinanceAnalystRole → GetObject on S3 bucket/2023/*
|   |--- Azure: Power-BI Viewer + Refresh rights on dataset
|
|--- PAM (on-demand) → Temporary admin for 30 min if approved
```

Ask a cybersecurity question...

--- PAM (on-demand) → Temporary admin for 30 min if approved

Take-away

- Identify the exact business tasks → translate them into the smallest set of permissions.
- Create dedicated groups/roles (AD, IAM, DB) that embody those minimal permissions.
- Assign users only to those groups, never directly to broad admin groups.
- Use time-bound or just-in-time elevation for any occasional privileged work.
- Automate audits and monitoring to ensure the "least-privilege" state is maintained over time.

Applying PoLP in this structured way lets the organization protect its critical data, limit the impact of credential compromise, and stay compliant with standards such as ISO 27001, NIST 800-53, and GDPR, while still enabling the finance analyst to do her job efficiently.

Ask a cybersecurity question...